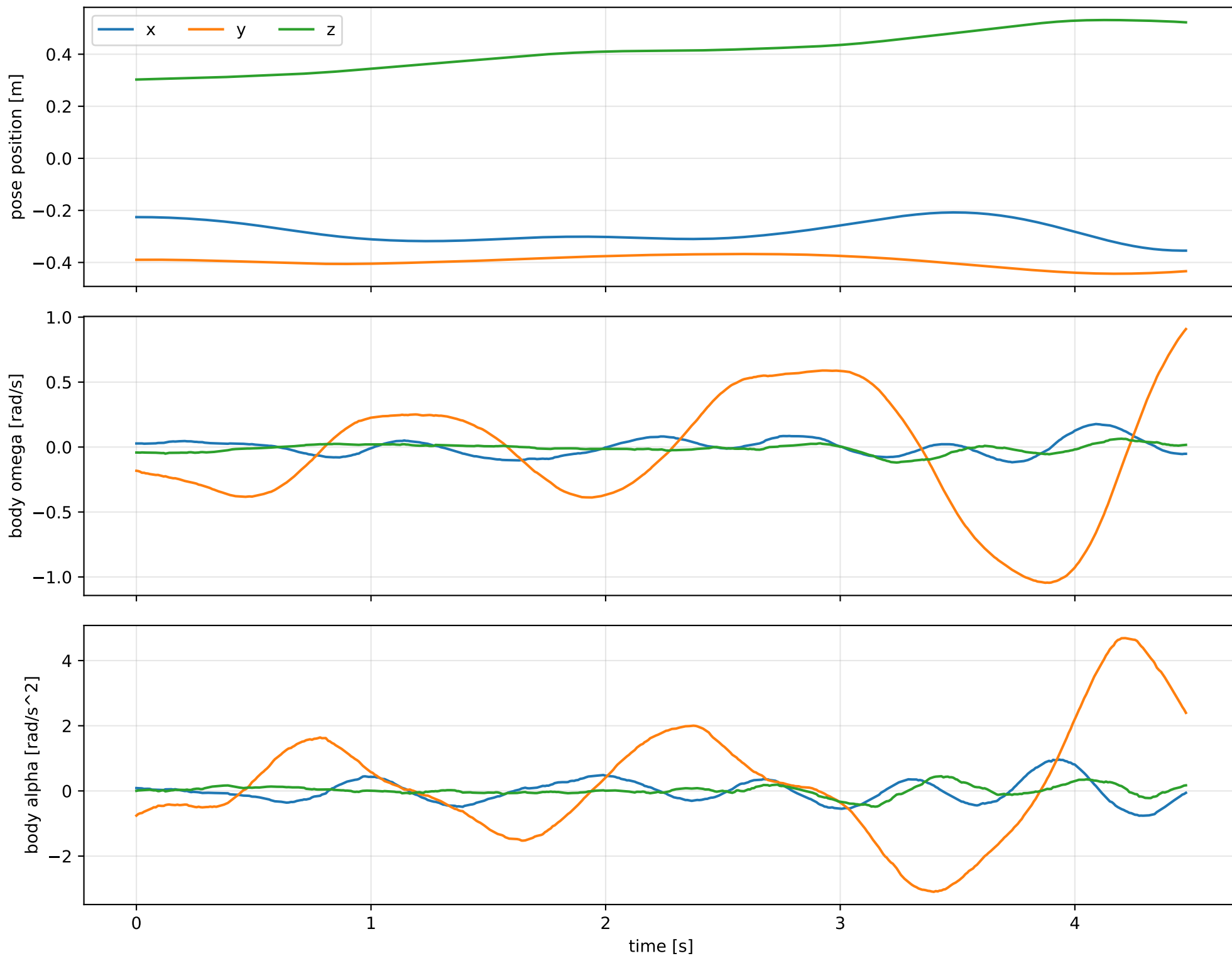
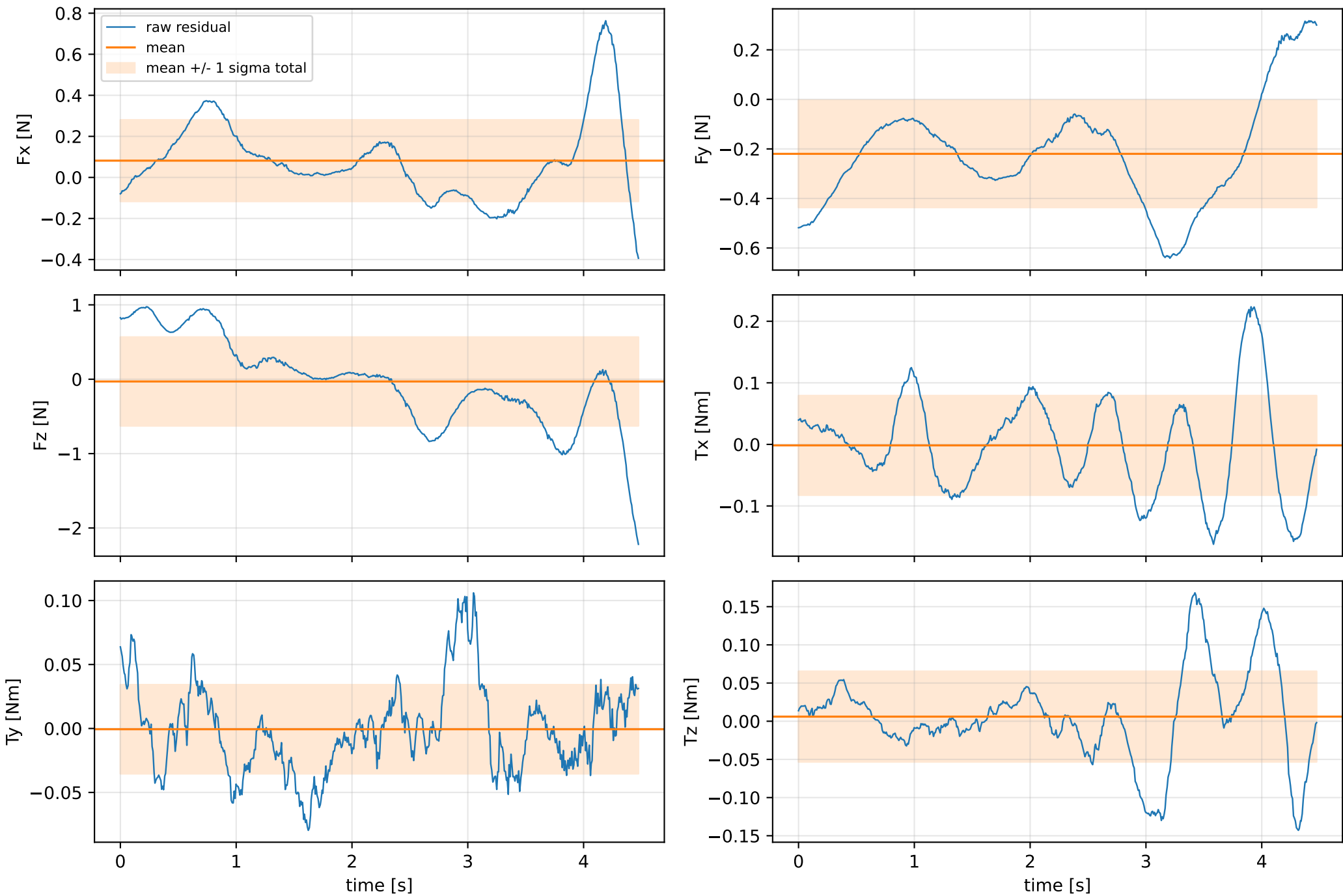


cog_x_nominal: SG trajectory and derived motion



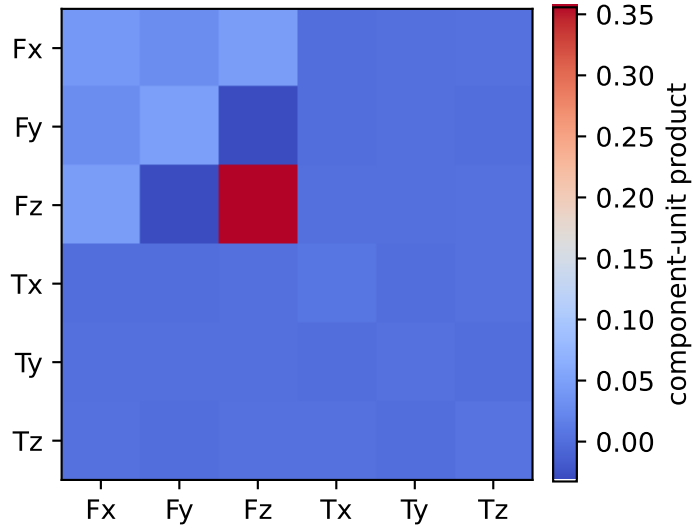
cog_x_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



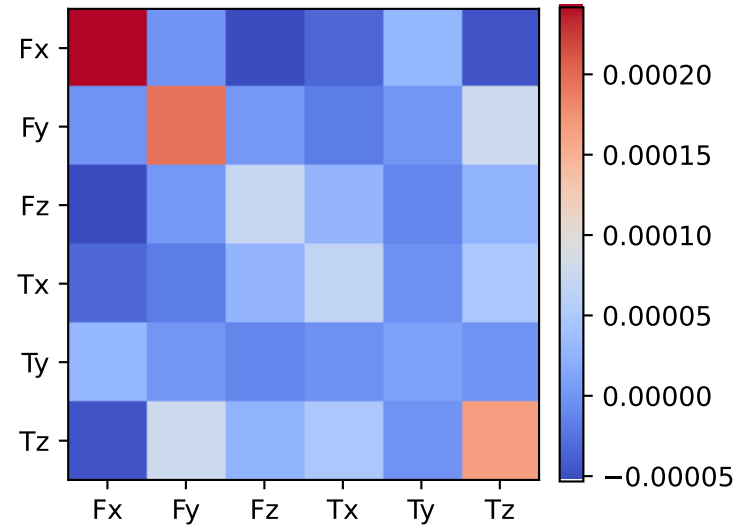
vector RMS about zero: force 0.70497 N, torque 0.10613 Nm; component std about mean: force [0.1984 0.2164 0.5964], torque [0.0808 0.0348 0.0593]

cog_x_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.007 0.008 0.07 0.364]

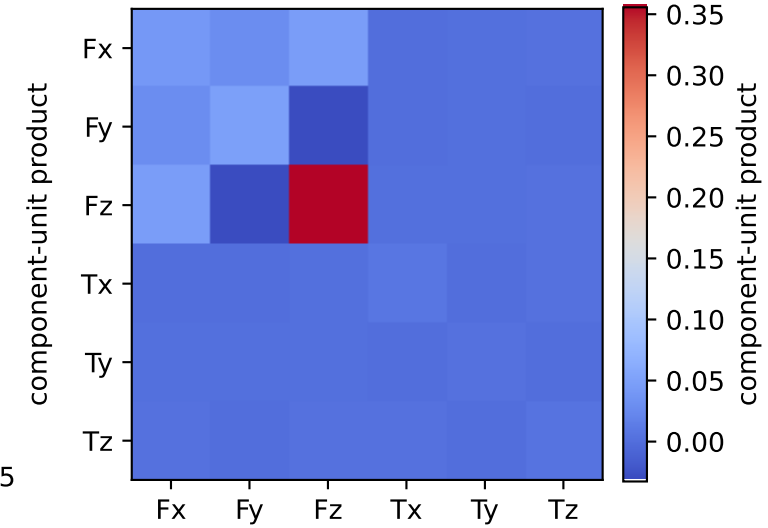
empirical total covariance



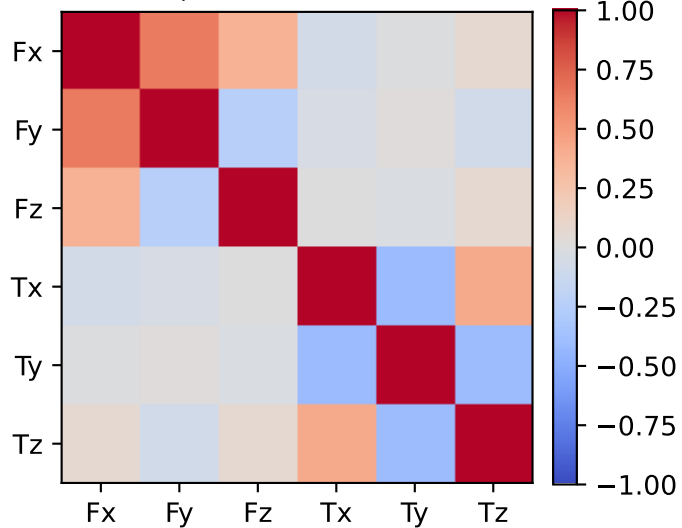
mean full-SG prediction covariance



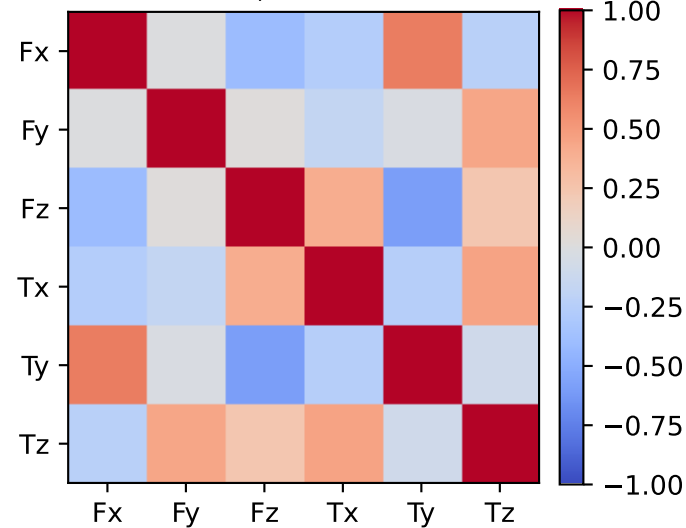
PSD excess model discrepancy covariance



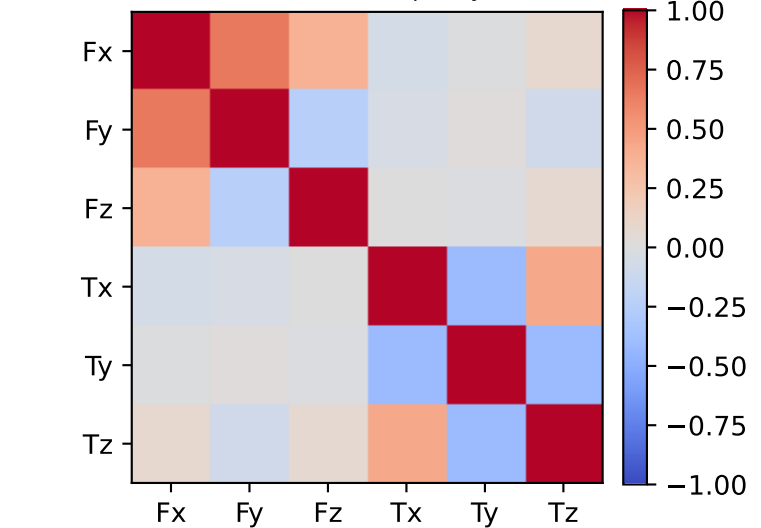
empirical total correlation



mean full-SG prediction correlation

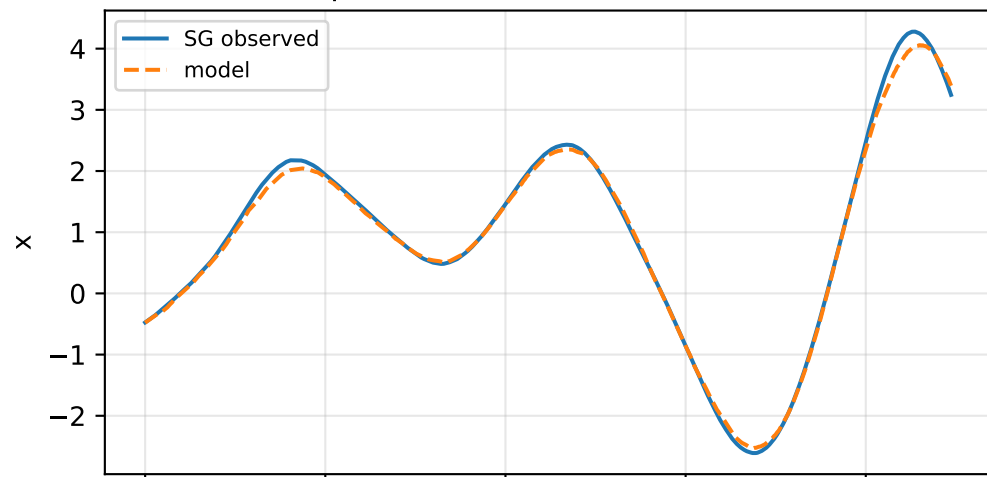


PSD excess model discrepancy correlation

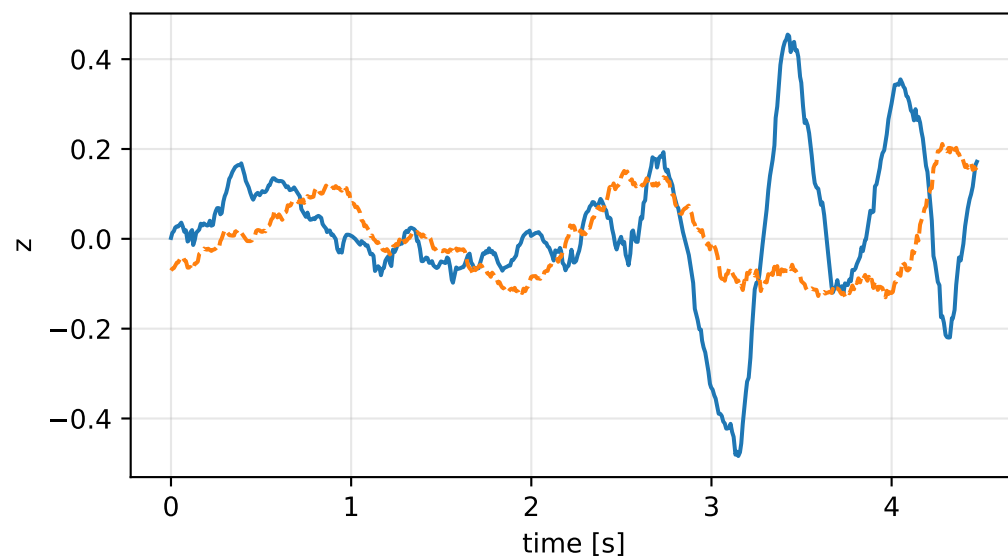
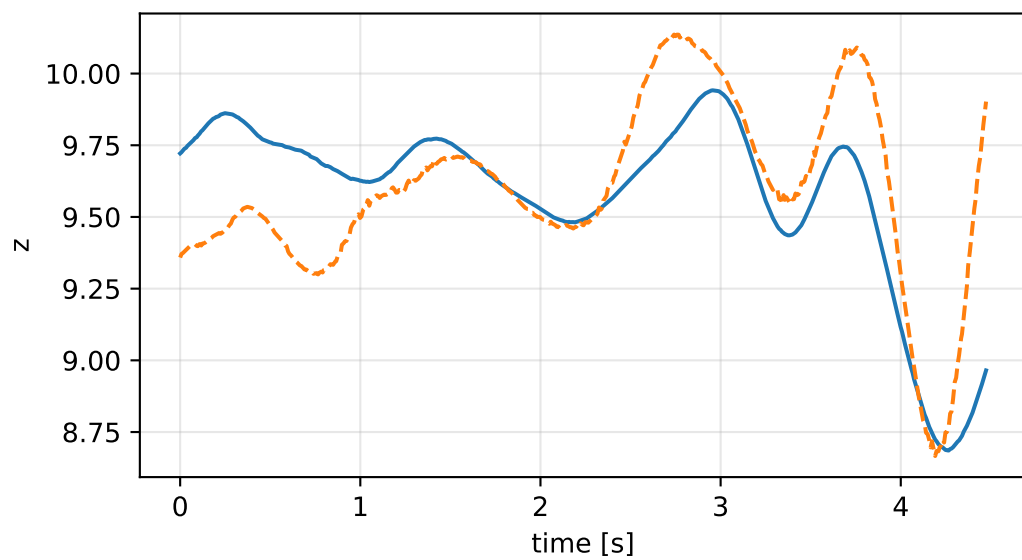
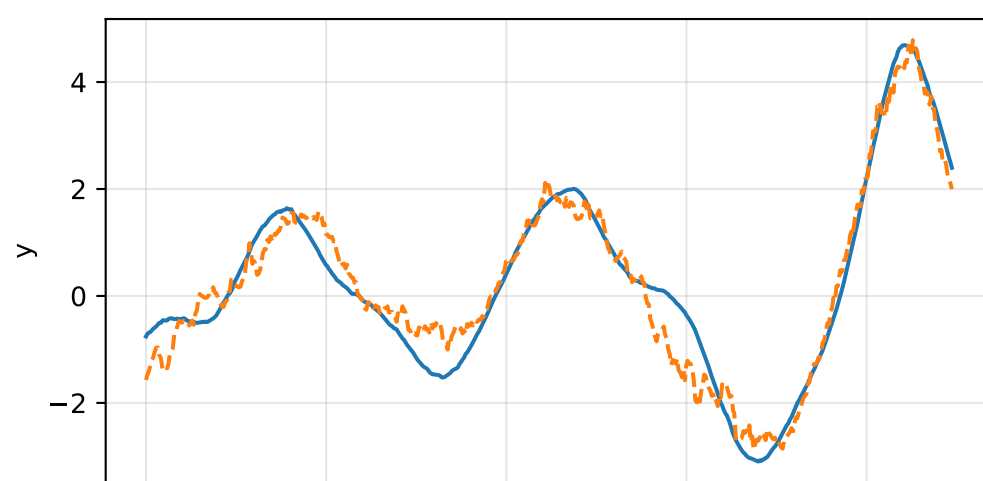
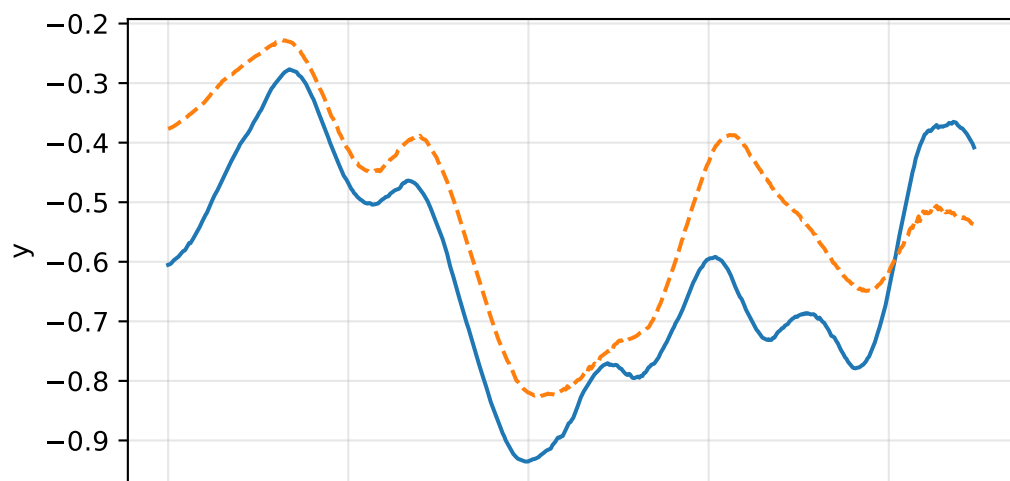
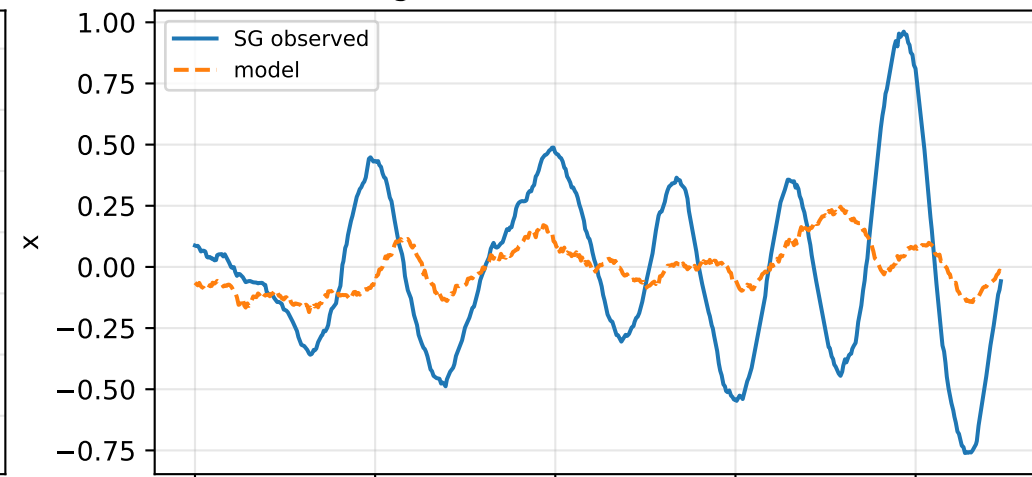


cog_x_nominal: acceleration residual objective

specific acceleration [m/s^2]

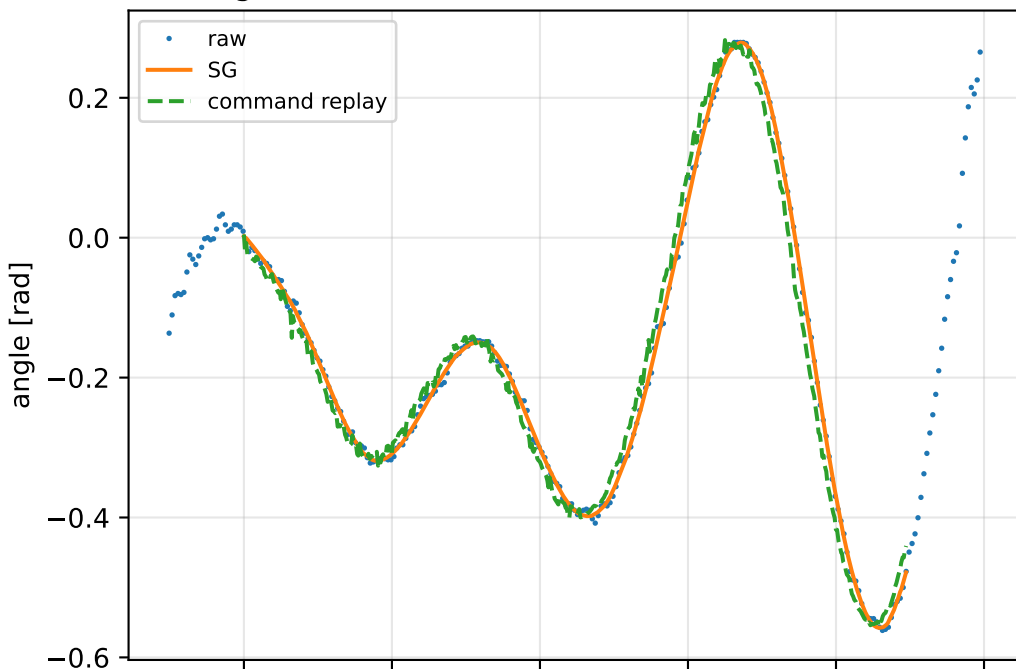


angular acceleration [rad/s^2]

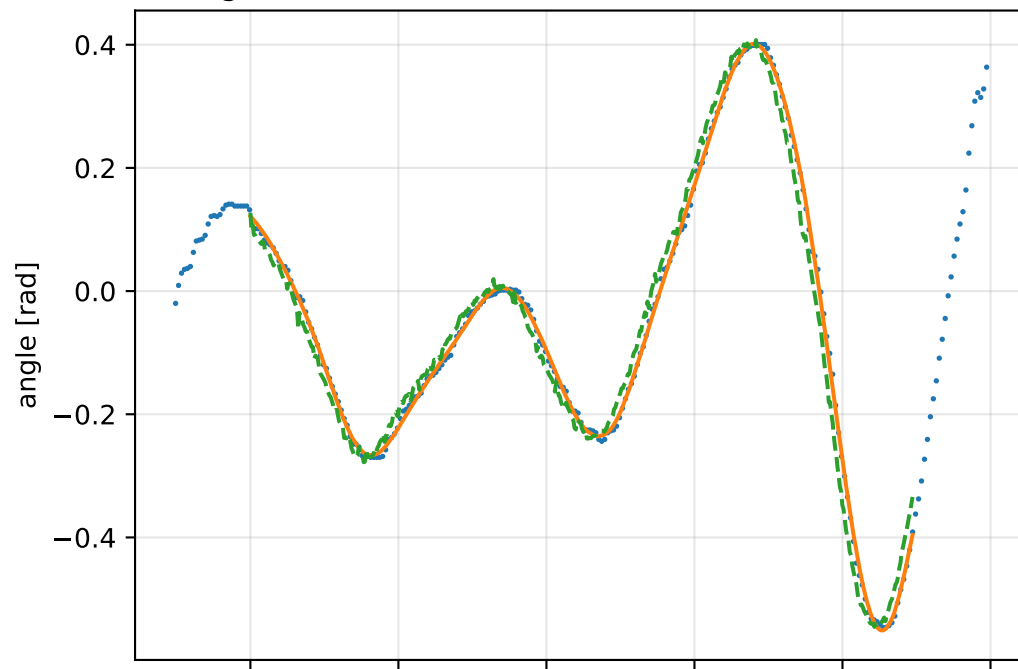


cog_x_nominal: gimbal measurement audit (objective=measured_sg)

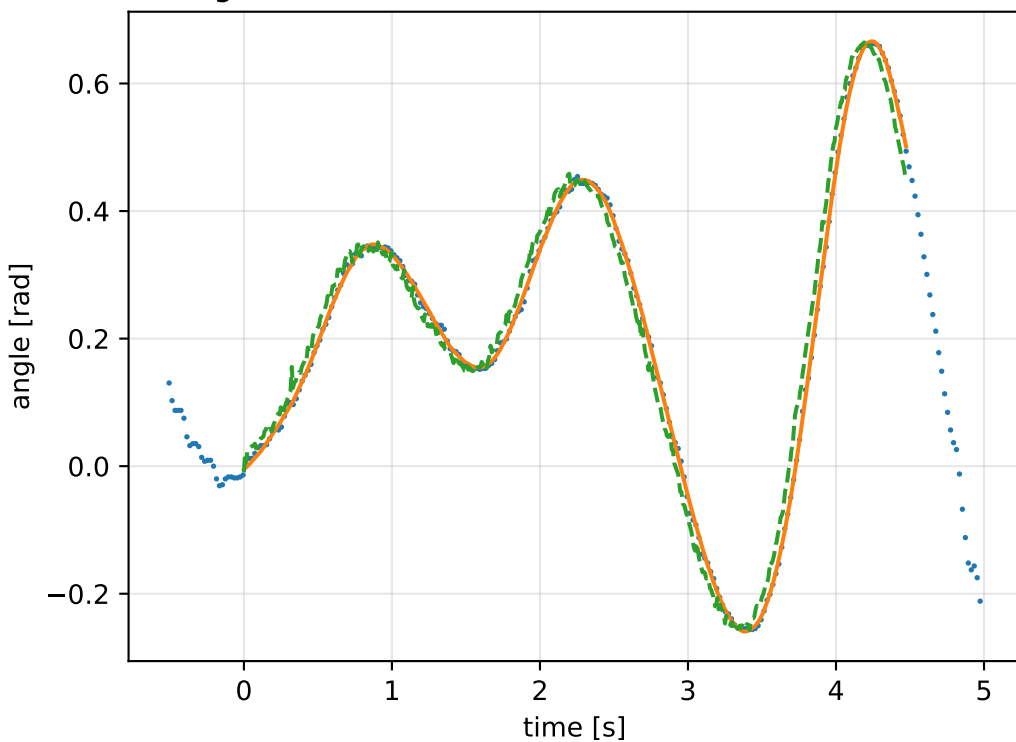
gimbal 1: RMSE=0.027, max=0.07529 rad



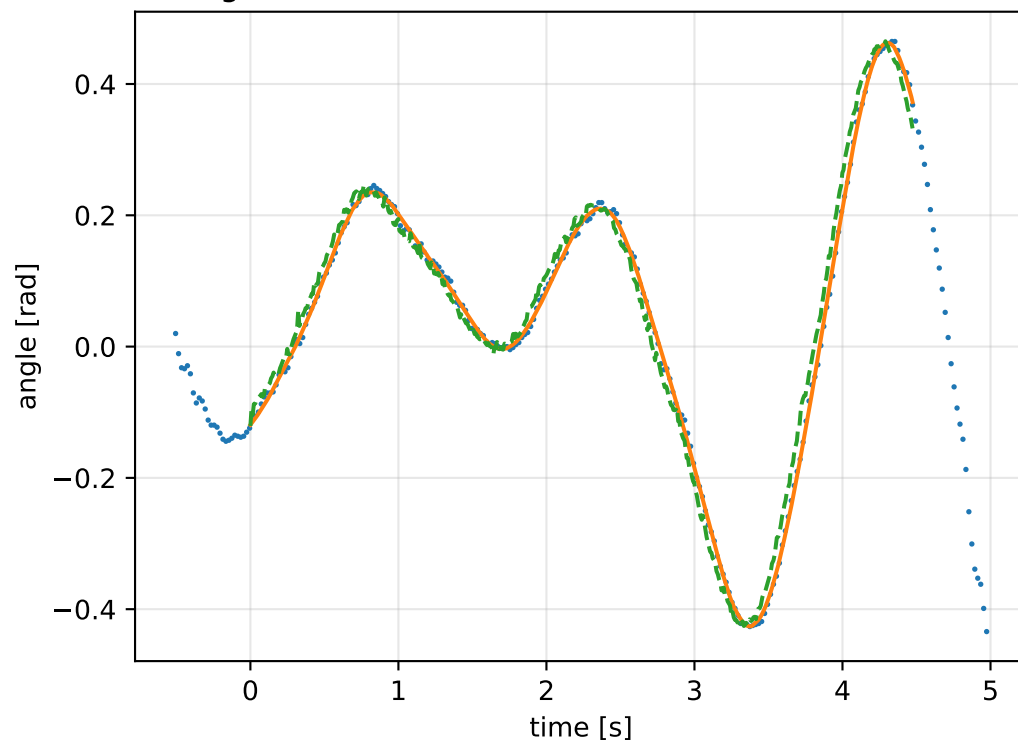
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

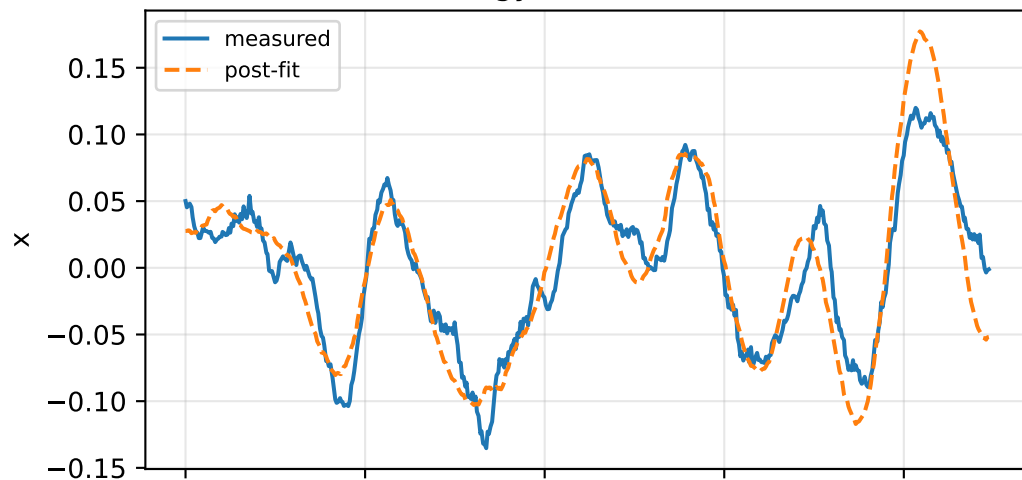


gimbal 4: RMSE=0.02598, max=0.07261 rad

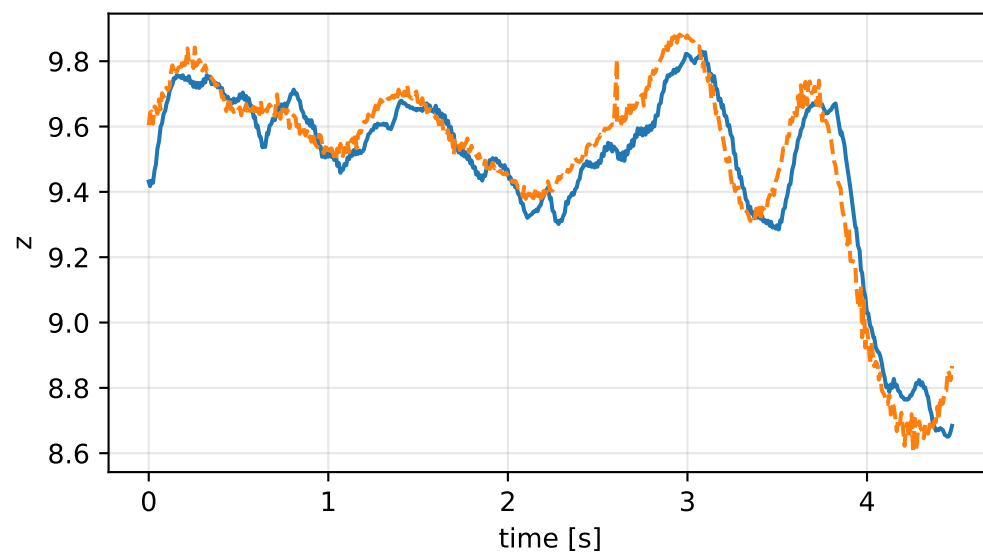
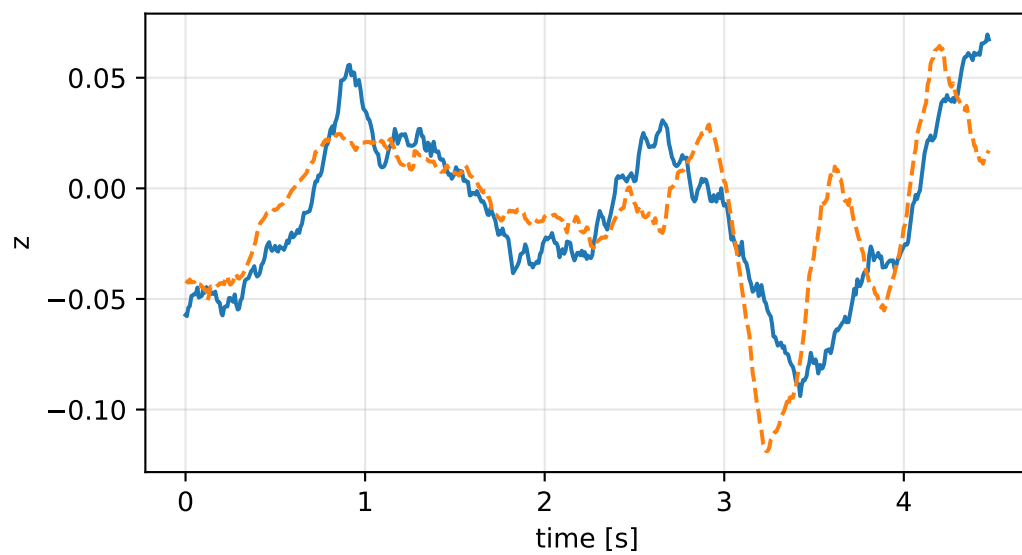
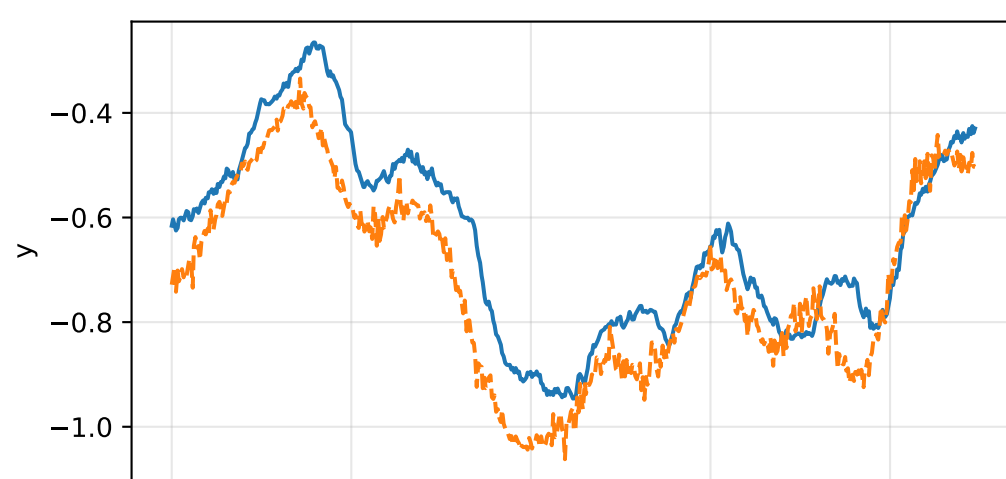
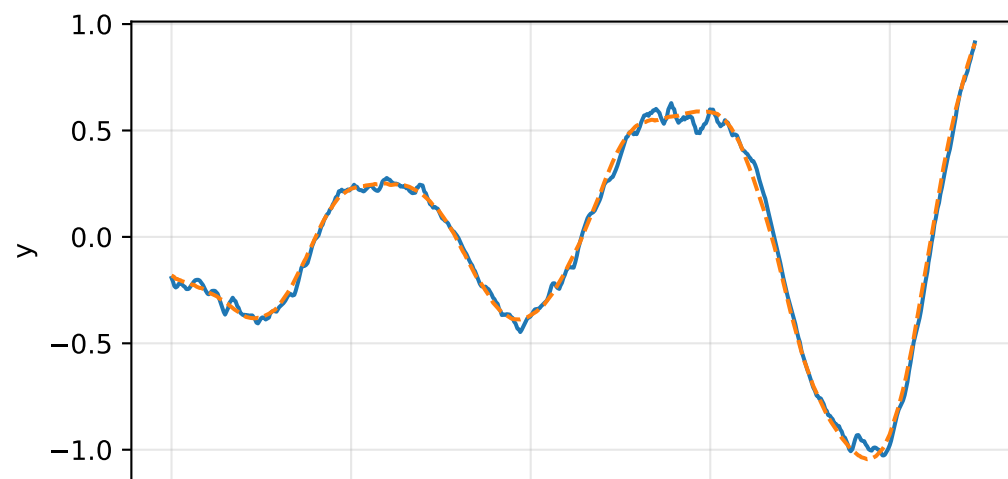
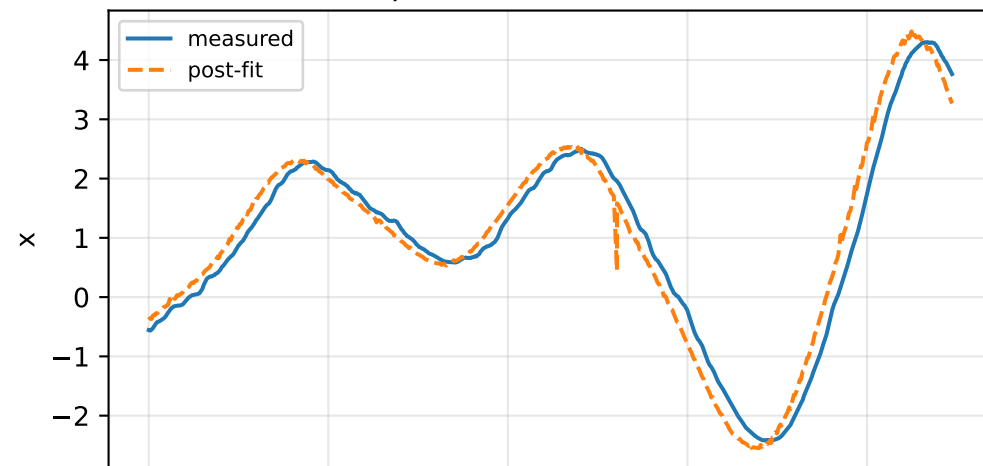


cog_x_nominal: IMU comparison (diagnostic only)

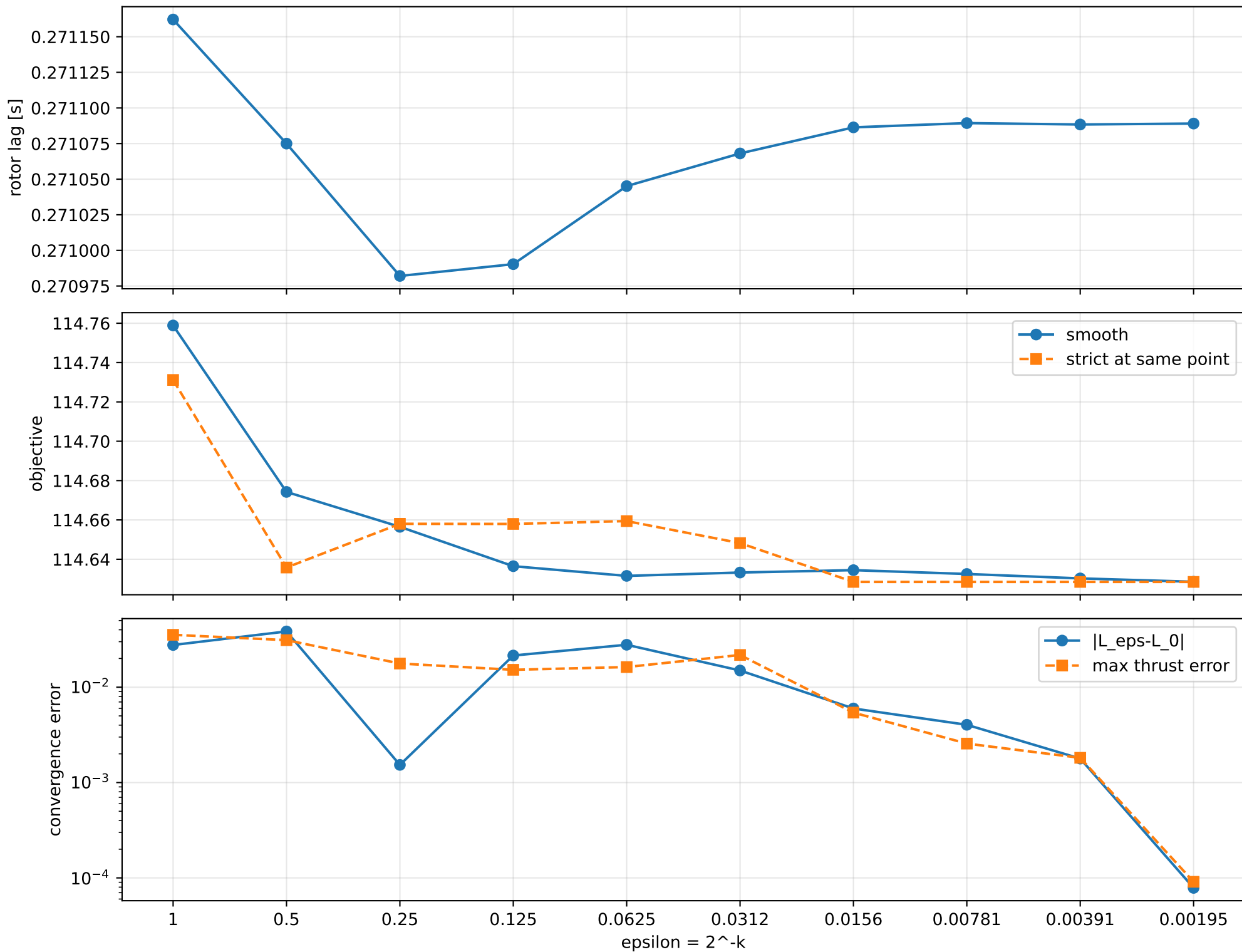
gyro [rad/s]



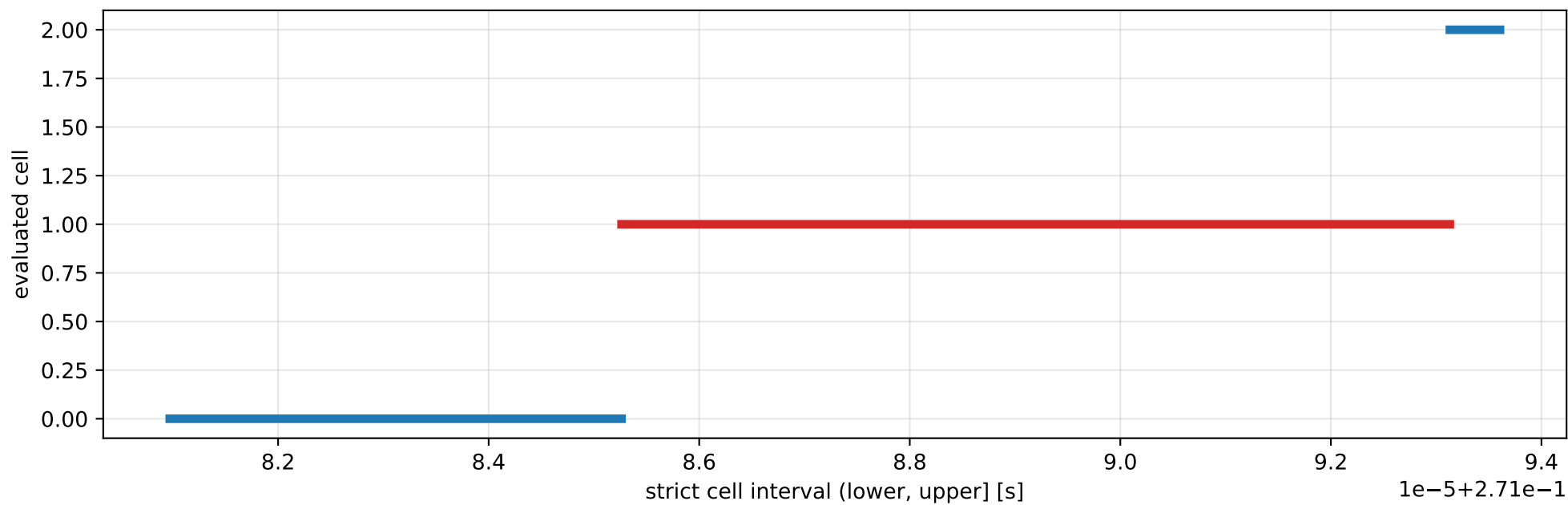
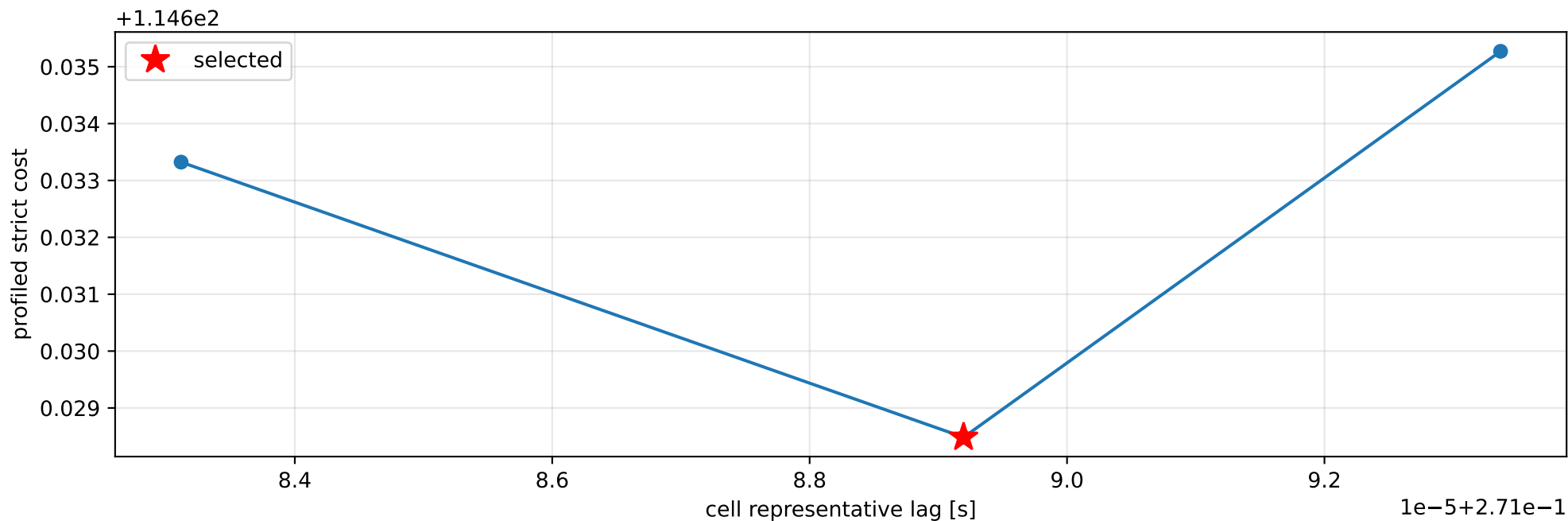
specific force [m/s^2]



cog_x_nominal: smooth-to-strict continuation



cog_x_nominal: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089049 s; data boundary=False

cog_x_nominal: parameter estimate

Case: cog_x_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.19414771

inertia [kg m²]:

```
[[ 0.42769826  0.00473833  0.02384566]
 [ 0.00473833  0.14093896 -0.02398309]
 [ 0.02384566 -0.02398309  0.55769564]]
```

CoG body [m]: [-0.00202472 0.00289709 0.01127322]

force effectiveness: [1.52050595 1.49690019 1.56752287 1.47914281]

scale-free J/m [m²]:

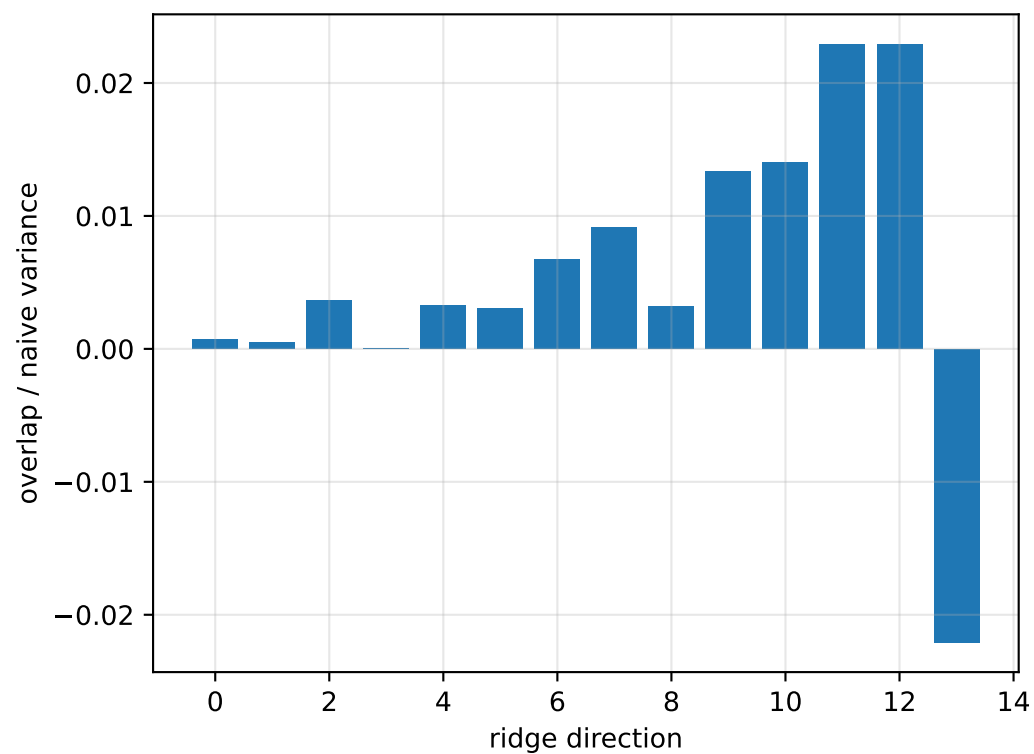
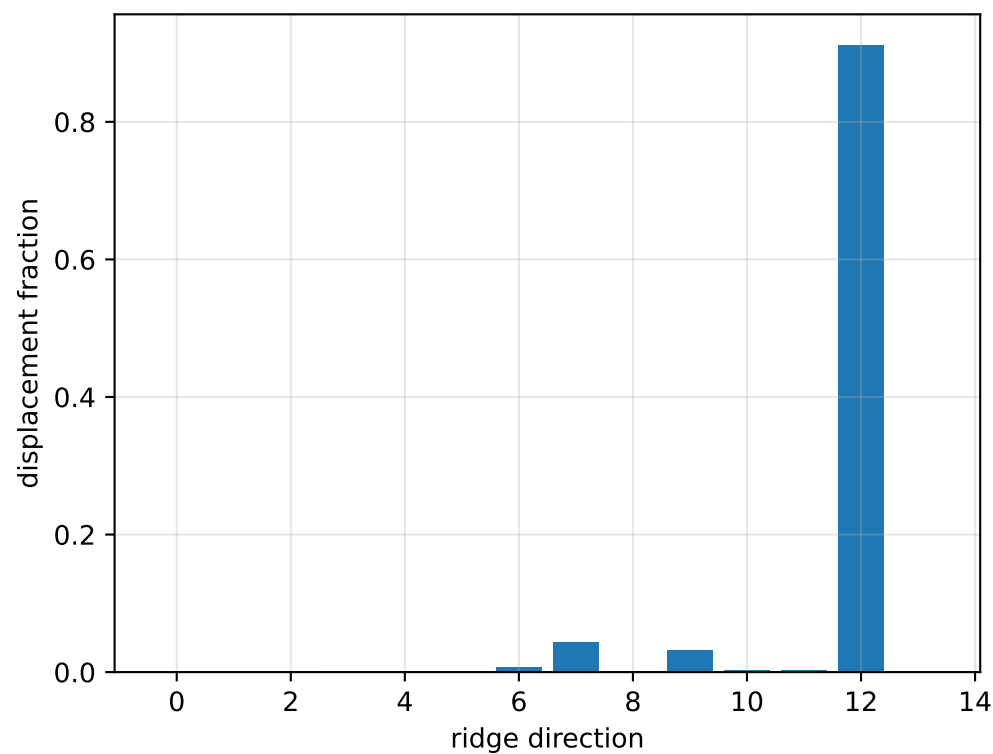
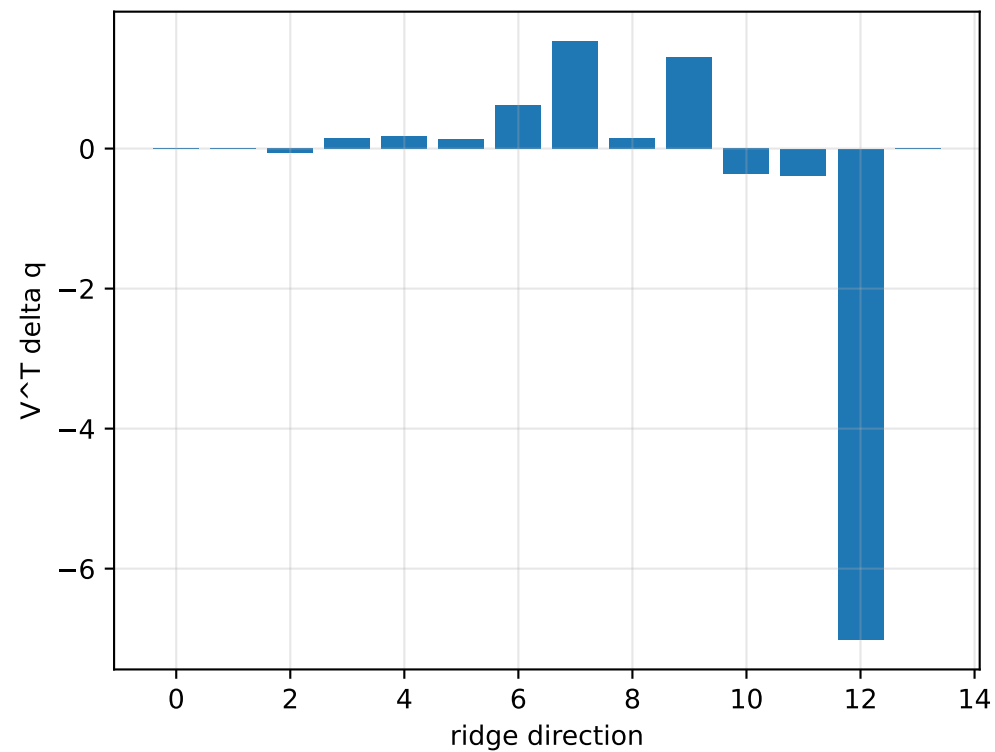
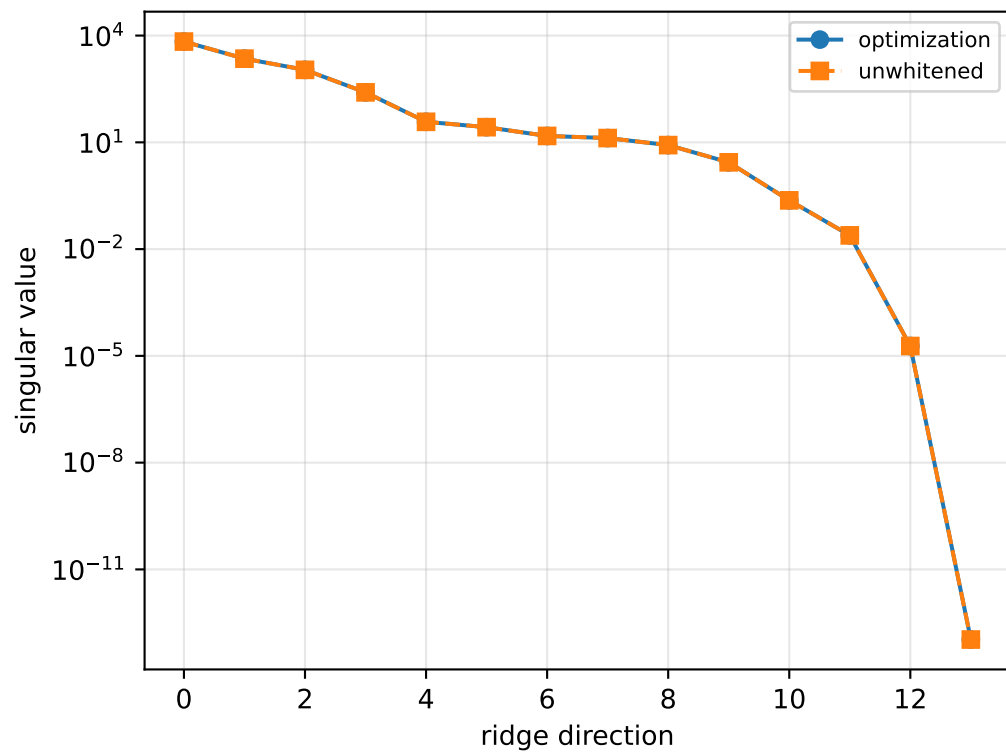
```
[[ 0.10197501  0.00112975  0.00568546]
 [ 0.00112975  0.03360372 -0.00571823]
 [ 0.00568546 -0.00571823  0.13296996]]
```

scale-free f/m: [0.36253038 0.35690212 0.3737405 0.35266827]

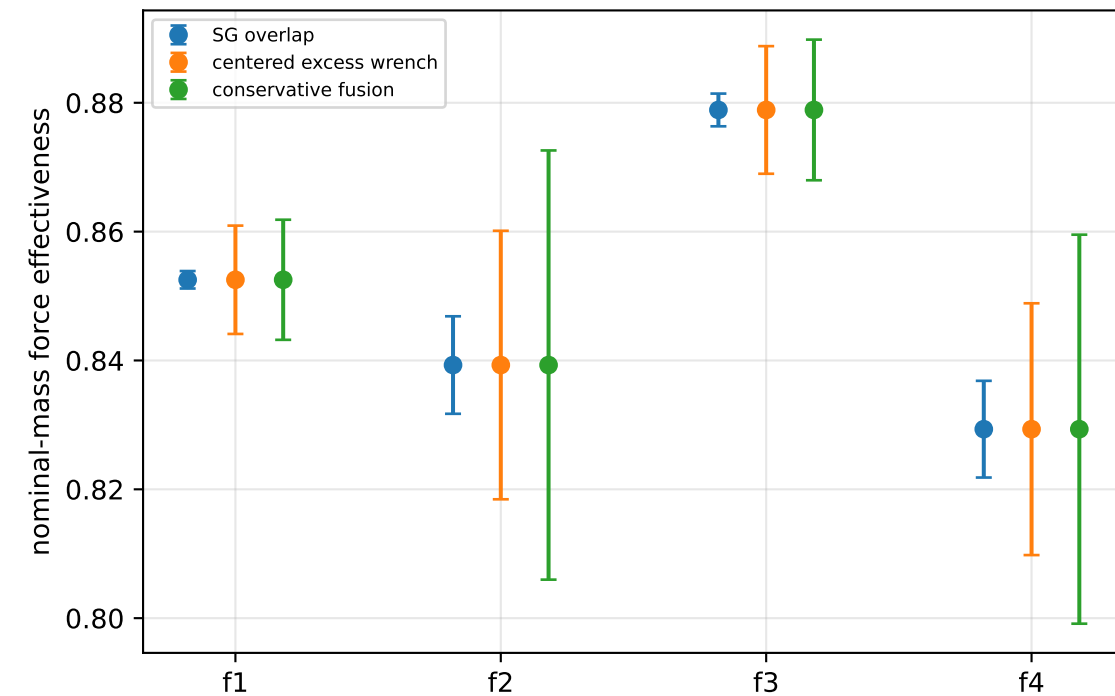
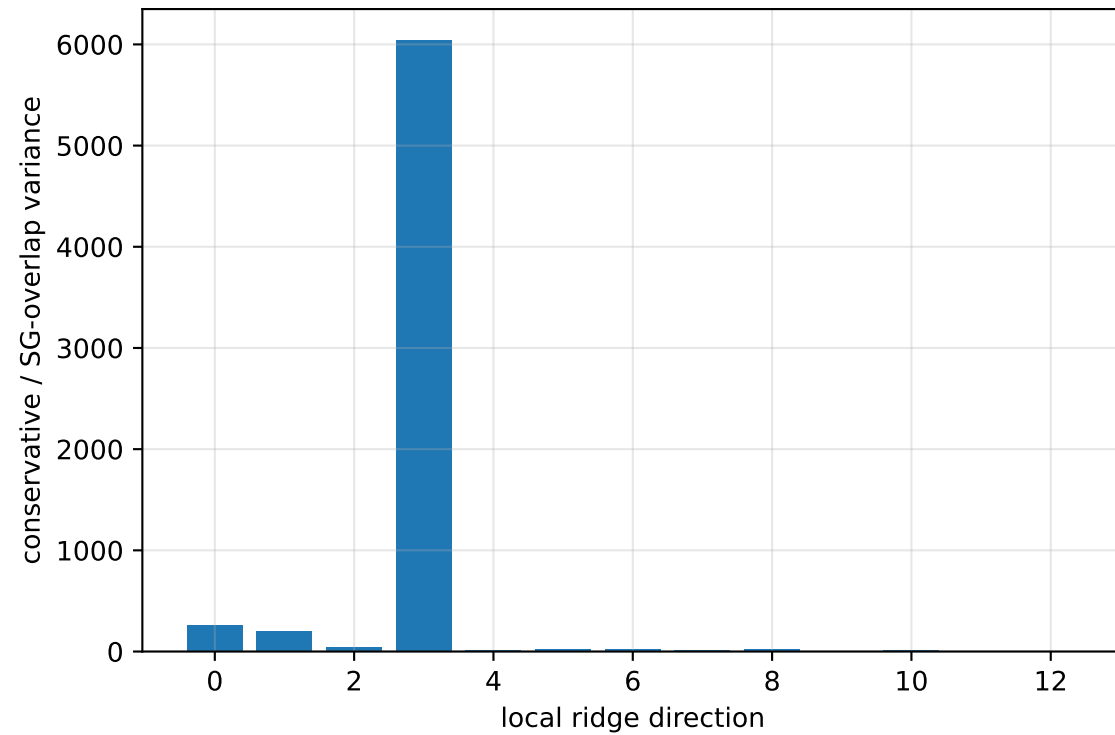
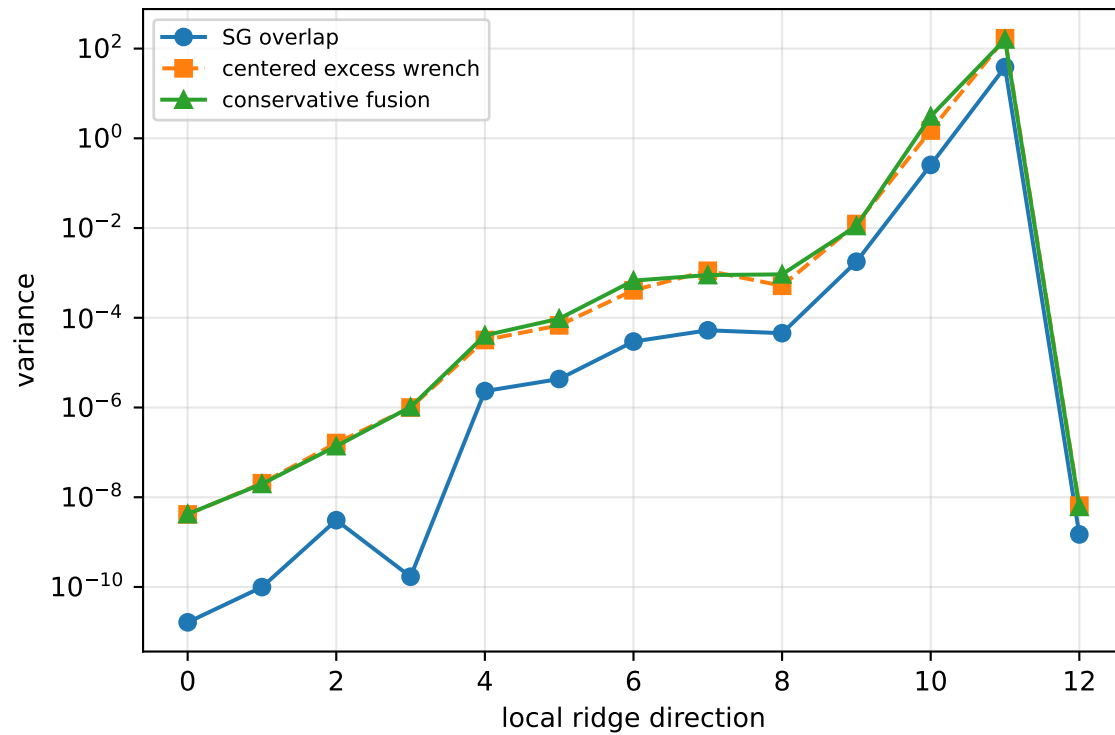
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

cog_x_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=1.64e-13)



cog_x_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 232$
wrench-to-acceleration recovery max error = $2.94\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

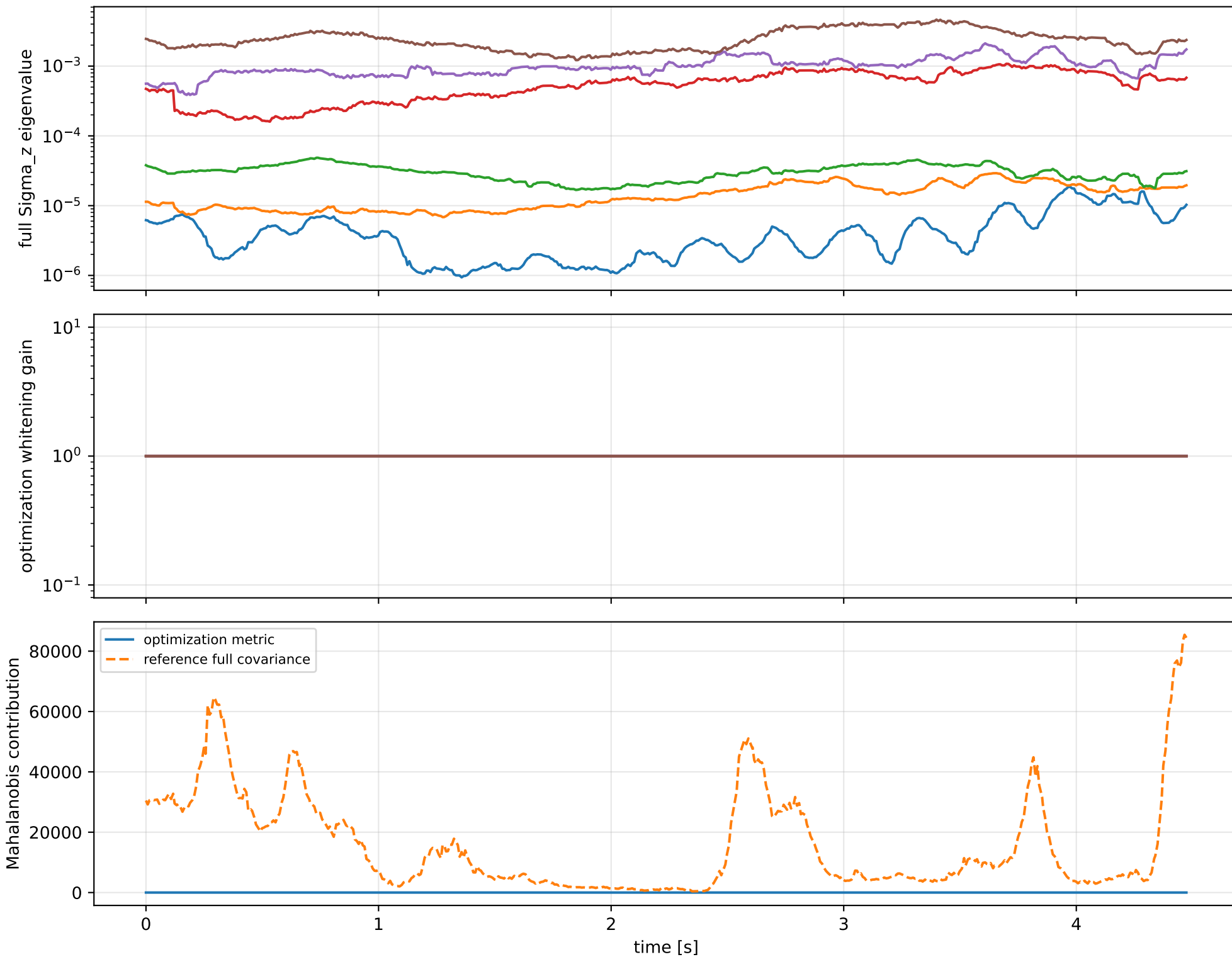
index 11: variance=159.5, ratio=4.117
 $\log_{\text{mass}}=+0.0369$; second_moment_diag_1=+0.554
second_moment_diag_2=-0.14; second_moment_diag_3=-0.598
second_moment_offdiag_12=+0.215; second_moment_offdiag_13=-0.282
second_moment_offdiag_23=+0.428; cog_x=+2.49e-05
cog_y=-1.94e-05; cog_z=-4.96e-07
log_force_eff_1=+0.0369; log_force_eff_2=+0.0371
log_force_eff_3=+0.037; log_force_eff_4=+0.0367

index 10: variance=3.083, ratio=12.08
 $\log_{\text{mass}}=+0.0228$; second_moment_diag_1=-0.0389
second_moment_diag_2=+0.36; second_moment_diag_3=-0.436
second_moment_offdiag_12=-0.638; second_moment_offdiag_13=-0.364
second_moment_offdiag_23=-0.37; cog_x=+0.00147
cog_y=-0.00188; cog_z=+9.12e-05
log_force_eff_1=+0.0237; log_force_eff_2=+0.0391
log_force_eff_3=+0.0205; log_force_eff_4=+0.00894

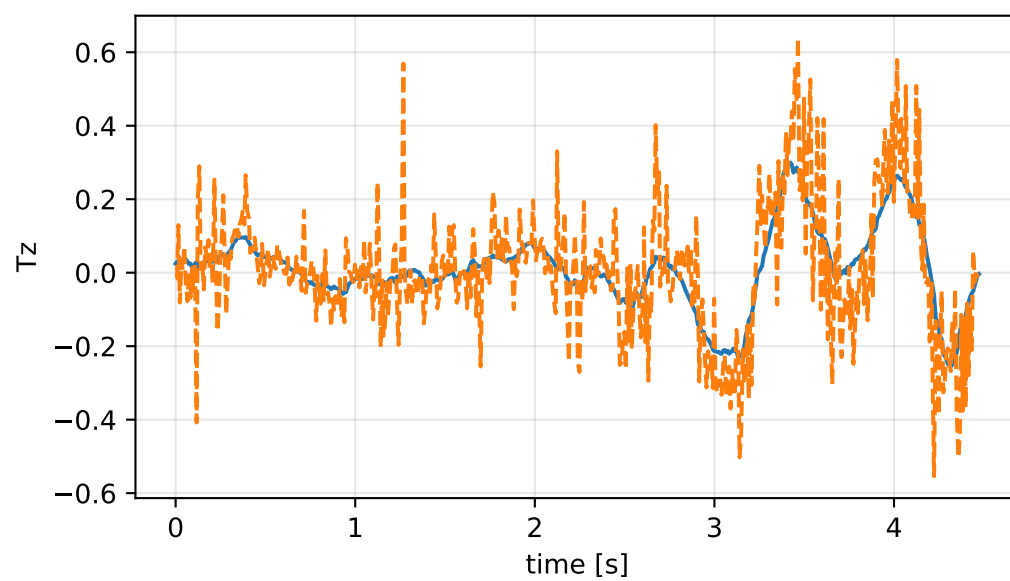
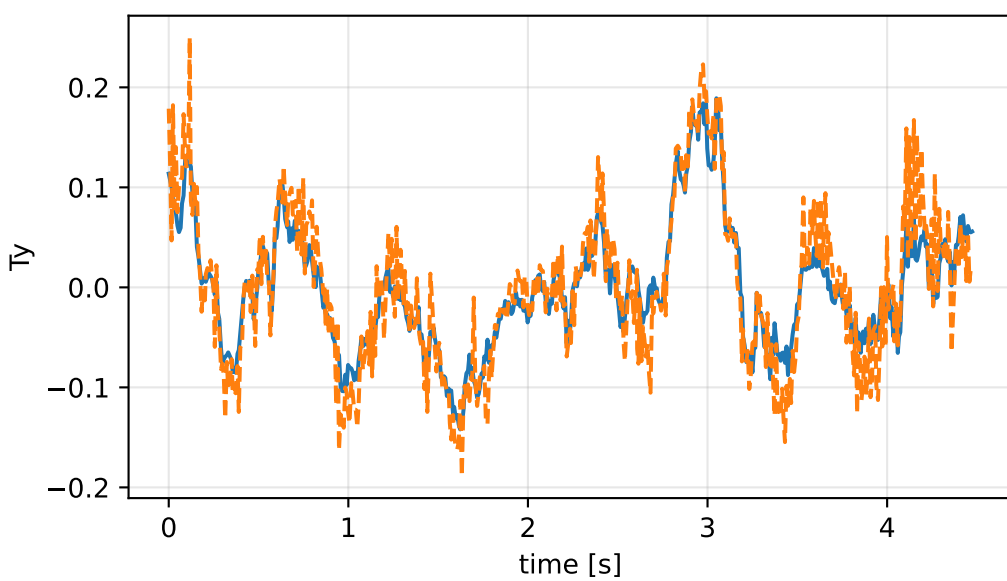
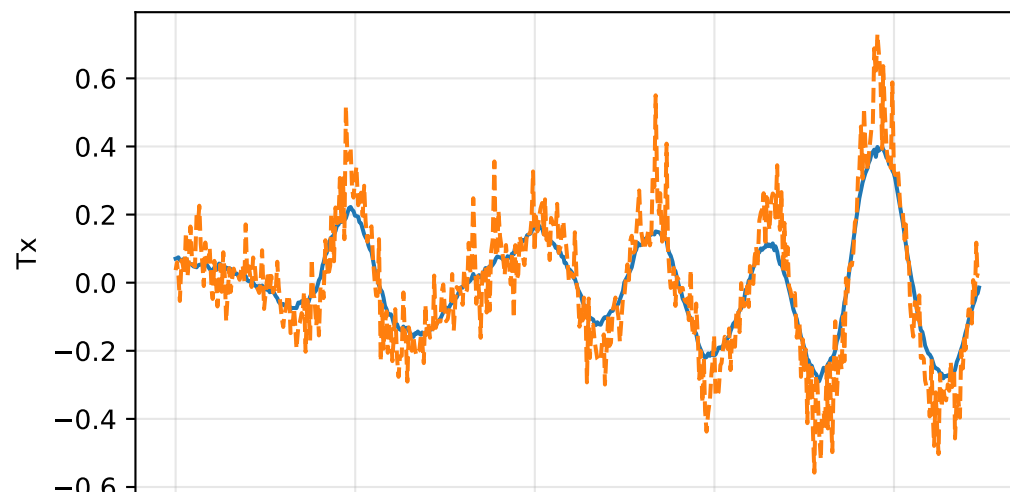
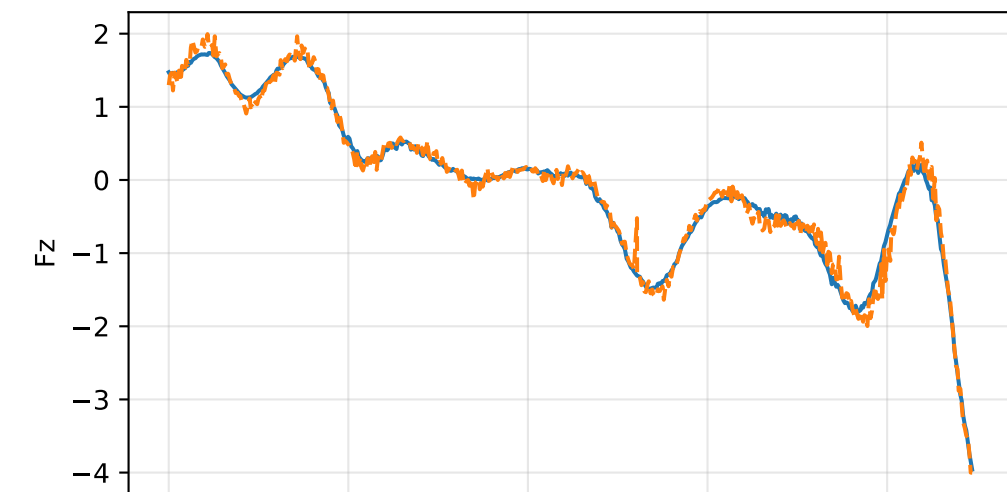
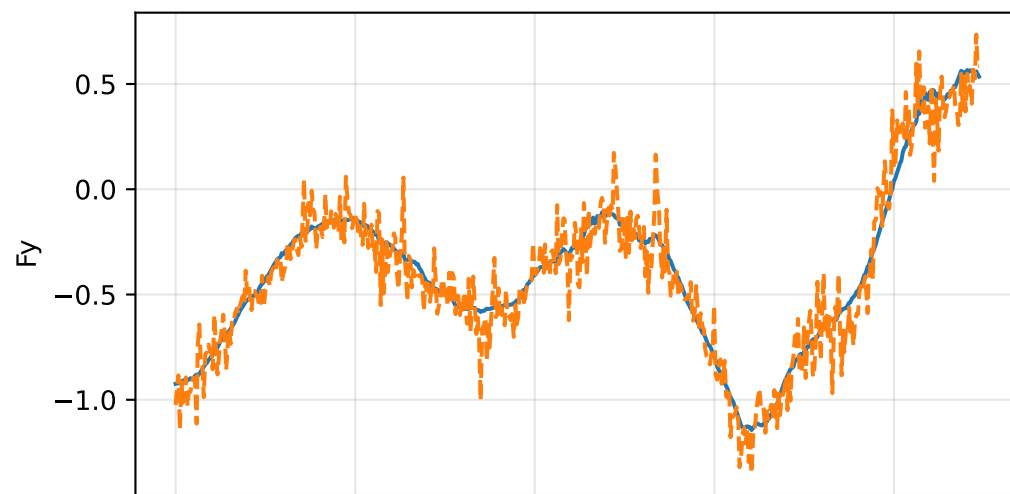
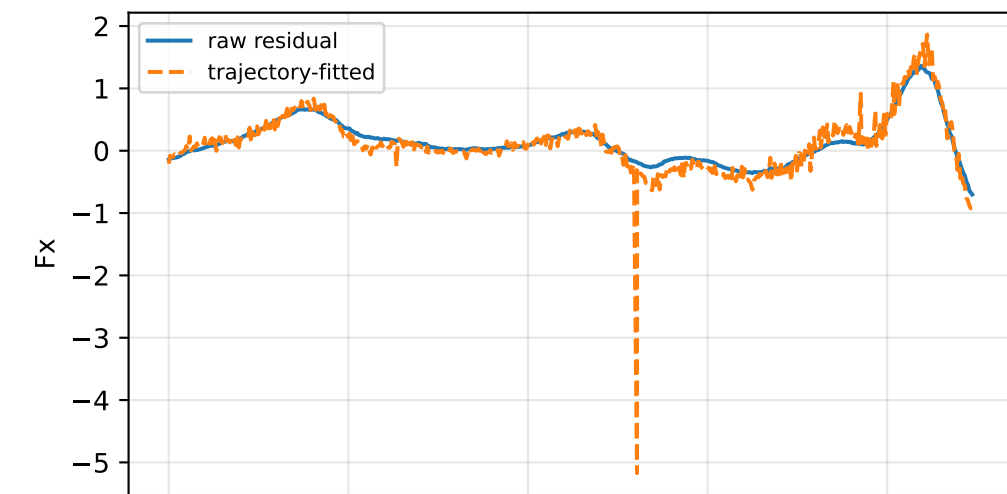
index 9: variance=0.01112, ratio=6.268
 $\log_{\text{mass}}=-0.16$; second_moment_diag_1=-0.031
second_moment_diag_2=+0.81; second_moment_diag_3=+0.0218
second_moment_offdiag_12=+0.094; second_moment_offdiag_13=+0.119
second_moment_offdiag_23=+0.436; cog_x=+0.00119
cog_y=+0.00214; cog_z=-0.00043
log_force_eff_1=-0.173; log_force_eff_2=-0.166
log_force_eff_3=-0.144; log_force_eff_4=-0.158

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_x_nominal: optimization vs reference covariance



cog_x_nominal: wrench / closure (force max 1.24e-14, torque max 1.6e-16)



cog_x_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_cog_x_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/cog_x_nominal.j

SHA256: bbfe2d2485bf27a6773ee0d77fa915418fa80f5a855431ac4e09df6ae9248b8d

data objective: 114.628483727

prior objective: 9.32033296097e-07

total objective: 114.628484659

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog_x_nominal (cog_position_body_m)

components: x

target: [-0.00202471]

std: [1.e-05]

final: [-0.00202472]

error: [-1.36530824e-08]

standardized residual: [-0.00136531]

factor objective: 9.32033296097e-07